Lecture 18: PCA and Linear Time Series Analysis

IEOR E4709: Statistic Analysis and Time Series
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1 Principal Component Analysis

1.1 Property of minimum mean-squared error of representation

It can be shown that $X = \sum_{i=1}^{p} \alpha_i v_i^\top$. To see why this is true, let us first consider the variance-covariance matrix $\Sigma = \frac{1}{n-1} X^\top X$. This is a symmetric matrix and thus can be *diagonalized*, i.e.,

$$\Sigma = V \Lambda V^\top, \tag{1}$$

where Λ is a diagonal matrix whose entries are the corresponding eigenvalues ($\Lambda_{1,1}$ is the largest eigenvalue, $\Lambda_{2,2}$ is the second largest eigenvalue, etc...), V is a matrix whose columns are the eigenvectors of Σ (the first column of V is the eigenvector corresponding to the largest eigenvalue, the second column of V is the eigenvector corresponding to the second largest eigenvalue etc...).

We also know from linear algebra that the matrix X admits a *singular value decomposition*:

$$X = USV^\top, \tag{2}$$

where U is a unitary matrix, i.e., $U^\top U = UU^\top = I_n$, where I_n is the n-dimensional identity matrix. S is the matrix of singular values. Hence, using Eq. (2), we obtain

$$\Sigma = \frac{1}{n-1} X^\top X = \frac{1}{n-1} VSU^\top USV^\top = V \frac{S^2}{n-1} V^\top.$$

Hence, the eigenvalue λ_i of the covariance matrix is equal to $\frac{S_{i,i}^2}{n-1}$. Recall the matrix of principal components is given by $XV = USV^\top V = US$, where we have used that the eigenvectors of the symmetric matrix Σ are orthogonal. We know that the i-th column of the matrix XV is $(XV)^i = \alpha_i$. It then follows from Eq. (2) that X can be represented as $X = \sum_{i=1}^{p} \alpha_i v_i^\top$.

The so called "rank-r approximation" of the data matrix X in terms of the first r principal components is given by

$$X_r := \sum_{i=1}^{r} \alpha_i v_i^\top.$$

It is well known that X_r is the linear rank-r approximation of X that minimizes the Frobenius norm over all matrices of rank at most r (this is the Eckart–Young–Mirsky theorem whose proof can be

seen in linear algebra textbooks). We take this fact for granted, and compute the approximation error next as

$$\begin{aligned}
\frac{1}{n}||X - X_r||_F^2 &= \frac{1}{n}|| \sum_{i=r+1}^{p} \alpha_i v_i^\top ||_F^2 \\
&= \frac{1}{n} tr \left(\left(\sum_{i=r+1}^{p} \alpha_i v_i^\top \right) \left(\sum_{i=r+1}^{p} \alpha_i v_i^\top \right)^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i,j=r+1}^{p} \alpha_i v_i^\top v_j \alpha_j^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i=r+1}^{p} \alpha_i \alpha_i^\top \right) \\
&= \frac{1}{n} tr \left(\sum_{i=r+1}^{p} \alpha_i^\top \alpha_i \right) \\
&= \frac{1}{n} \sum_{i=r+1}^{p} ||\alpha_i||^2 = \frac{n-1}{n} \sum_{i=r+1}^{p} \lambda_i,
\end{aligned}$$

where $||A||_F^2 = \sum_{i,j} A_{ij}^2$ is the Frobenius norm of the matrix A, i.e. the sum of the absolute squares of its elements, while $||x||^2 = \sum_i x_i^2$ is the L_2 norm of the vector x. In the second equality of the derivation above, we have used the well known result that, for a matrix A, we have that $||A||_F = \sqrt{tr(AA^\top)}$. In the fourth equality, we have used that v_i and v_j are orthonormal. In the fifth equality, we have used that $tr(AB) = tr(BA)$. The above result has implications for data compression.

Remark 1.1. *The rank r approximation is not intended in the sense that the matrix X_r has rank r, but rather that the matrix has rank of at most r. X_r is the approximation of the data matrix X, which only makes use of the first r principal components, i.e. the principal components that have the r largest variances.*

1.2 Lack of Scale Invariance

Consider a transformation of the data, i.e., $\tilde{X} = XQ$. For example, if Q is a diagonal matrix, this means that we are re-scaling the units of the observation variables. Under a transformation Q, the transformed covariance matrix becomes $\tilde{\Sigma} = \frac{Q^\top X^\top XQ}{n-1} = Q^\top \Sigma Q$. Next, we want to ask the question: when do we have invariance of the PCA, i.e. under which transformation of the data, we have that $\lambda_i = \tilde{\lambda}_i$ and $\tilde{v}_i = Q^\top v_i$, i.e. the principal component vectors transform the same way as the data and the captured variances are the same? We are using the notation $\tilde{\Sigma}\tilde{v}_i = \tilde{\lambda}_i \tilde{v}_i$. For the scale invariance property to hold, we must guarantee that

$$\tilde{\lambda}_i = \tilde{v}_i^\top \tilde{\Sigma} \tilde{v}_i = v_i^\top \Sigma v_i = \lambda_i,$$

where $\tilde{\Sigma} = Q^\top \Sigma Q$ and $\tilde{v}_i = Q^\top v_i$. Plugging this expression in the equation above, we obtain that $v_i^\top QQ^\top \Sigma QQ^\top v_i = v_i^\top \Sigma v_i$, which holds true if $QQ^\top = I$, so the scale invariance property is maintained only under orthogonal transformations of the data.

Remark 1.2. *It is usually a good idea to standardize the data, i.e. demand each column of the data matrix, and divide each column by the standard deviation. The reason is that if some variables have a large variance and some others a small variance, the PCA will place higher loading on the columns with large variances. For example, if you change one variable from kilometers to meters (thus increasing its variance), it may go from having small impact to dominating the first principal component. To make the PCA independent of such rescaling, standardizing the variables is a good approach to follow. If we standardize the data, the covariance matrix of the data becomes the correlation matrix.*

2 Principal Component Regression

Recall that the variance of the least square estimator $\hat{\beta}$ in linear regression is $\sigma^2(X^\top X)^{-1}$. If X is column-rank deficient, meaning that one or more of its columns is equal to a linear combination of the others, then it means that one predictor variable is redundant. We refer to this problem as *collinearity*. In that case, the matrix $(X^\top X)$ would be non-invertible. Even if there is not an exact linear relationship among the predictors, but one predictor is approximately a linear combination of the others, then we would have that the inverse matrix $(X^\top X)^{-1}$ would be huge, and thus the variance of the least square estimator $\hat{\beta}$ would be very large. This happens, for instance, if we have many predictor variables compared to the number of observations (i.e., $p >> n$). In such case, one possibility is to reduce the dimensionality of the data matrix X first, which is at the heart of principal component regression.

The key idea is that often a small number of principal components is sufficient to explain most of the variability of the data, as well as the relationship with the response. Hence, we modify the multiple linear regression model from $y = X\beta + \epsilon$ to $y = X'\gamma + \epsilon$. where the matrix $X' = (\alpha_1, \alpha_2, \ldots, \alpha_r)$ is the matrix whose first column is the principal component α_1, the second column is the principal component α_2, ..., and the r-th column is the principal component α_r.

Remark 2.1. *Let us call the least square estimate of the principal component regression $\hat{\gamma}$. We can recover the coefficient β associated with the regression using the original data matrix X, i.e., the linear regression model $y = X\beta + \epsilon$, from the coefficient vector γ. To do so, we use the following relation*

$$\beta = V\gamma, \tag{3}$$

where $V = (v_1, ..., v_r)$ and v_i is the i-th eigenvector. To see why the relation (3) *holds, observe the following: $y = X'\gamma + \epsilon = \sum_{i=1}^{r} \gamma_i \alpha_i + \epsilon = \sum_{i=1}^{r} X v_i \gamma_i + \epsilon = XV\gamma + \epsilon = X\beta + \epsilon$, where the last equality holds true if $\beta = V\gamma$. If the standard PCA and PCR packages of R are used, one gets the principal components α_i and the least square estimates $\hat{\gamma}$.*